
GLA UNIVERSITY, MATHURA

Department of Computer Engineering & Applications



PROJECT SYNOPSIS

TOPIC: MINI PROJECT SYNOPSIS ON

JobBridge — Job Portal (MERN Stack)

B.Tech. — Computer Science & Engineering

Submitted By

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Submitted To

Mr. Akash Kumar Choudhary
Technical Trainer

Academic Session: 2026–27

DECLARATION.

We hereby declare that the proposed mini project entitled —JobBridge — Job Portal (MERN Stack) is our original academic work and will be designed and developed by the undersigned students as part of the B.Tech. Computer Science & Engineering curriculum. The project will follow clean coding practices, REST API design principles, and Git/GitHub based version control.

The work presented in this synopsis is prepared for academic purposes. All members will contribute to the analysis, design, development, testing, documentation, and presentation of the project.

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Project Mentor / Technical Trainer: Mr. Akash Kumar Choudhary

ACKNOWLEDGEMENT.

We express our sincere gratitude to Mr. Akash Kumar Choudhary, Technical Trainer, GLA University, for his guidance and valuable suggestions during the planning of this mini project. His direction has helped us understand the expected structure, technical scope, and development approach.

We also thank the Department of Computer Engineering & Applications and GLA University for providing an academic environment in which we can apply concepts of web development, databases, software engineering, and version control to a practical problem.

Finally, we appreciate the cooperation of all team members. The project will be completed through coordinated work involving requirement analysis, UI development, server-side programming, database design, API integration, testing, documentation, and final presentation.

Project: JobBridge — Job Portal (MERN Stack)

Team: Milind Singh, Abhishek Gupta, Vedant Bansal, Rohit Chaudhary, Ravindra Kumar Gautam

CERTIFICATE.

This is to certify that the students listed below are working on the mini project titled —JobBridge — Job Portal (MERN Stack) under the guidance of Mr. Akash Kumar Choudhary, Technical Trainer, GLA University.

The project synopsis has been prepared to describe the proposed problem, objectives, scope, technology stack, working methodology, modules, data flow, requirements, and planned deliverables of the system.

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Structured reference for the project synopsis.

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The synopsis follows the structure supplied for the mini-project proposal and presents the project in a clear academic format suitable for review and print submission.

1. INTRODUCTION

Project Title: JobBridge – Job Portal (MERN Stack)

About the Project

JobBridge is a full-stack web application designed to connect job seekers with employers through a centralized online platform. Employers can publish job opportunities and manage applications, while job seekers can discover relevant jobs, view details, and submit applications.

The system is planned using the MERN stack – MongoDB for data storage, Express.js and Node.js for server-side APIs and business logic, and React.js for an interactive client interface. RESTful APIs provide communication between the frontend and backend.

Problem Statement

Traditional job searching may involve scattered sources, repeated manual communication, and limited application tracking. Employers also need a simple mechanism to publish openings and review candidates. JobBridge aims to provide one organized platform for these activities.

Expected Outcome

A responsive and maintainable job portal with secure authentication, role-based functionality, job listing management, application handling, search/filtering, and application-status tracking.

2. PRIMARY REASON TO CHOOSE THIS PROJECT

The project is selected because employment platforms are a practical application of modern web technologies and demonstrate important full-stack concepts in one system. It provides opportunities to implement frontend design, REST APIs, authentication, database relationships, validation, testing, and Git-based collaboration.

3. THE MAIN OBJECTIVE OF THE PROJECT

- Develop a centralized platform for job seekers and employers.
- Allow employers to create, update, delete, and manage job listings.
- Allow job seekers to search, filter, view and apply for jobs.
- Maintain application records and application status.
- Implement authentication and role-based authorization.
- Design clean REST APIs and a maintainable MERN architecture.

Literature / Existing-System Survey

Existing online recruitment systems show that searchable job listings, user profiles, employer dashboards, and application tracking are core features of a useful recruitment platform. JobBridge adopts these common functional ideas while keeping the proposed system focused on a clear academic MERN implementation.

4. SCOPE OF THE PROJECT

- Job Seekers: registration, login, profile details, job search, filtering, job details, applications and status tracking.
- Employers: registration, company/profile details, job posting, editing/removal of listings and applicant management.
- Application Management: application creation, retrieval and status updates such as Pending, Accepted or Rejected.
- Administration: the architecture can be extended later with moderation, analytics and reporting features.

WORKING METHODOLOGY OF THE PROJECT

Phase	Planned Work
1. Requirements	Identify users, workflows, functional and non-functional requirements.
2. Design	Prepare UI structure, database schema, API contracts and DFDs.
3. Development	Build React frontend, Node/Express backend and MongoDB data layer.
4. Integration	Connect frontend components with REST APIs and authentication.
5. Testing	Perform functional, API, validation and basic security testing.
6. Deployment & Review	Deploy the application, verify workflows and document the final system.

5. DETAILS ABOUT THE HARDWARE AND THE SOFTWARE

SYSTEM REQUIREMENTS

Category	Requirement
Operating System	Windows 10/11, Linux or macOS
Runtime	Node.js and npm
Database	MongoDB / MongoDB Atlas
Browser	Chrome, Edge, Firefox or equivalent modern browser
Development Tools	VS Code or equivalent IDE; Git and GitHub
Testing	Browser DevTools and API testing with Postman/Thunder Client

HARDWARE REQUIREMENTS

A laptop/desktop with a modern multi-core processor, at least 8 GB RAM, adequate SSD/HDD storage, keyboard/mouse, and a stable internet connection for package installation, database access and Git/GitHub collaboration.

FRONTEND AND BACKEND

Frontend: React.js, HTML5, CSS3 and JavaScript. React components will provide responsive pages, forms, dashboards, search/filter interfaces and application views.

Backend: Node.js with Express.js will expose REST APIs, validate requests, implement authentication/authorization, manage business logic and communicate with MongoDB.

6. MODULE DESCRIPTION

Authentication & Users: Registration, login, password protection, JWT-based sessions and role-based access.

Job Management: Create, view, update and delete job postings; search and filter available jobs.

Applications: Apply to jobs, prevent invalid submissions, view submitted applications and track status.

Employer Dashboard: Manage posted jobs and review/update applicant statuses.

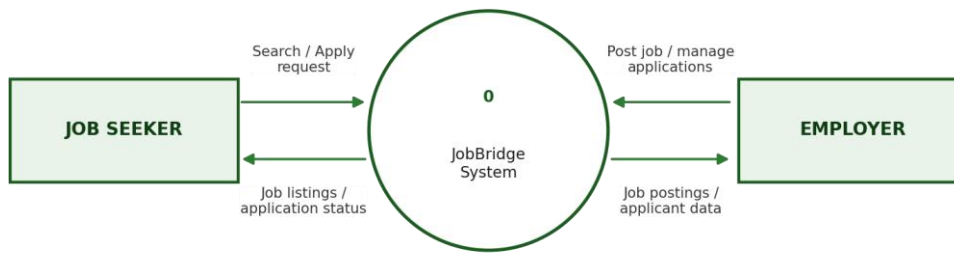
Database Layer: Store users, jobs and applications in MongoDB with appropriate references.

DATA FLOW DIAGRAMS

A DFD illustrates how data is processed by the system in terms of inputs, outputs, processing and storage, using standard notation: rectangles for external entities, circles for processes, open-ended rectangles for data stores and labelled arrows for data flow.

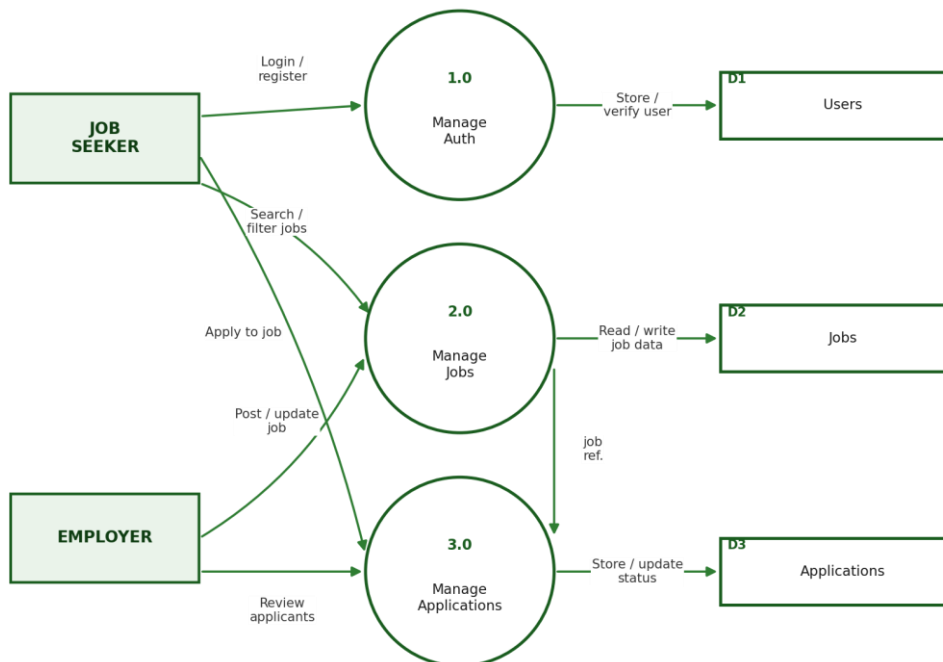
Level 0 DFD – Context Diagram

Level 0 DFD – Context Diagram



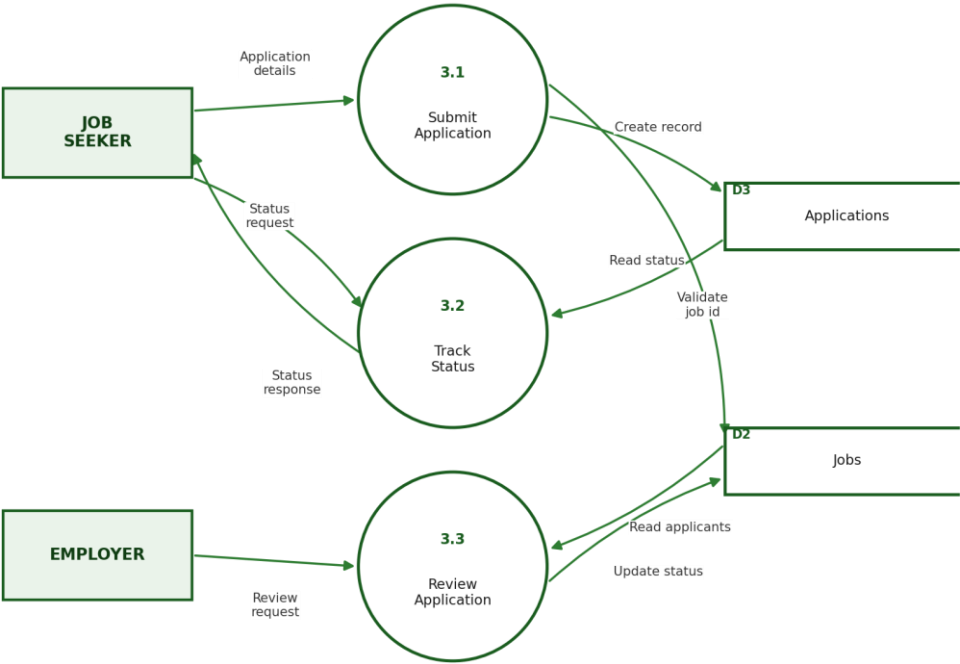
Level 1 DFD

Level 1 DFD



Level 2 DFD – Decomposition of Process 3.0 (Manage Applications)

Level 2 DFD — Decomposition of Process 3.0 (Manage Applications)



7. AVAILABILITY

JobBridge is planned as a web-based application that can be accessed through a modern web browser. During development, the application can be run locally on the development system.

For final demonstration, the project may be deployed on a suitable hosting platform if required. MongoDB Atlas or another suitable database hosting option may be used for the database.

The availability of the deployed system will depend on the selected hosting service, database service and internet connection.

BASIC DATA ENTITIES

- Users – stores login and basic profile information along with the user's role.
- Jobs – stores job title, description, skills, location and employer information.
- Applications – stores applicant, job, submission information and application status.

PLANNED DELIVERABLES

- Working React frontend.
- Node/Express backend connected with MongoDB.
- Basic authentication and role-based access.
- Job posting, searching and application workflow.
- Tested project build and documentation.

8. BIBLIOGRAPHY / REFERENCES

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Development Practices

- Use Git/GitHub for version control, collaboration and source-code history.
- Follow modular React components and separated backend routes/controllers/services.
- Validate user input and protect authenticated endpoints with authorization checks.
- Use clear naming, reusable code, error handling and API response conventions.

9. PROJECT PLANNING & DELIVERABLES

Timeline	Activities	Deliverable
Week 1	Requirement analysis, problem definition and project planning	Requirement specification
Week 2	UI wireframes, database schema and API design	System design + schema
Week 3	React frontend and authentication	Frontend modules
Week 4	Node/Express REST APIs and MongoDB integration	Backend + APIs
Week 5	Job listing and application modules	Integrated application
Week 6	Testing, bug fixing and validation	Tested build
Week 7	Deployment, documentation and presentation preparation	Final project package

SUMMARY

JobBridge proposes a practical MERN-based solution for connecting job seekers and employers. The planned system combines authentication, role-based access, job listing, search/filtering, application management, REST APIs and MongoDB persistence in one maintainable full-stack application. The project will be developed incrementally with testing and Git/GitHub version control throughout the implementation.

— End of Project Synopsis —